

SmartCart

Real-Time Price Intelligence Platform

SmartCart saves you money before you spend it
— powered by PySpark, FastAPI & AWS EC2.



Presented by Arushika(2310990301)



The Problem

Overpaying

Indians overpay ₹1000s buying products without comparing prices across stores.

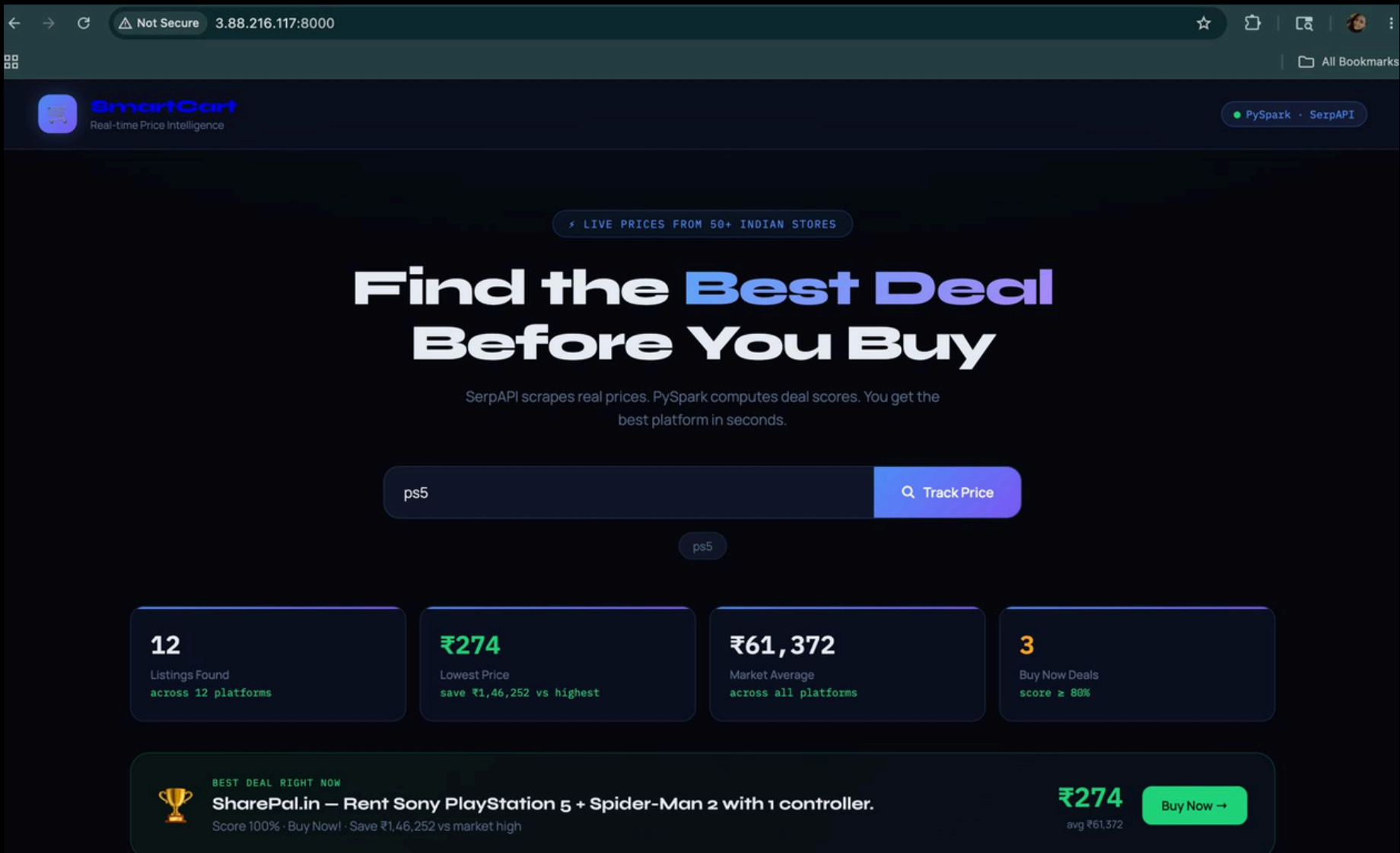
Manual Effort

Checking 10+ sites manually takes 30+ minutes — time most shoppers don't have.

Volatility

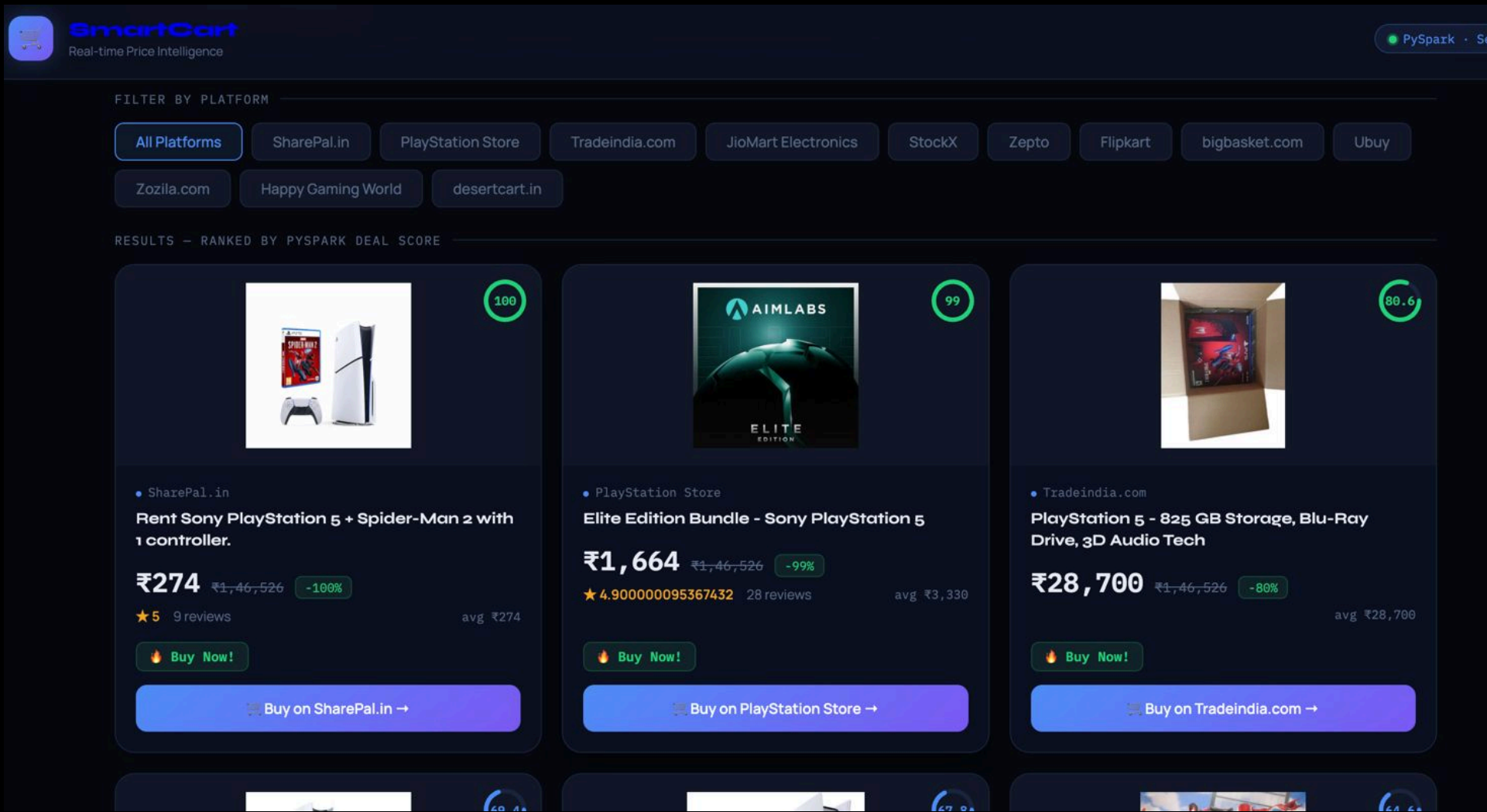
Prices change daily — no way to know the right time to buy.

Project Demo



Search Any Product

Enter any product name and get instant results from 50+ Indian stores.



PySpark Ranked Results

PySpark scores and ranks every listing with a Deal Score in seconds and also provides “Buy Now” or “Wait” recommendation

Price Comparison Chart

Visual chart comparing prices across platforms at a glance.



 **Savings Calculator**

How many units?

-

2

+

Best price: ₹274 × 2 units · Saving ₹2,92,504 total vs market high

₹548

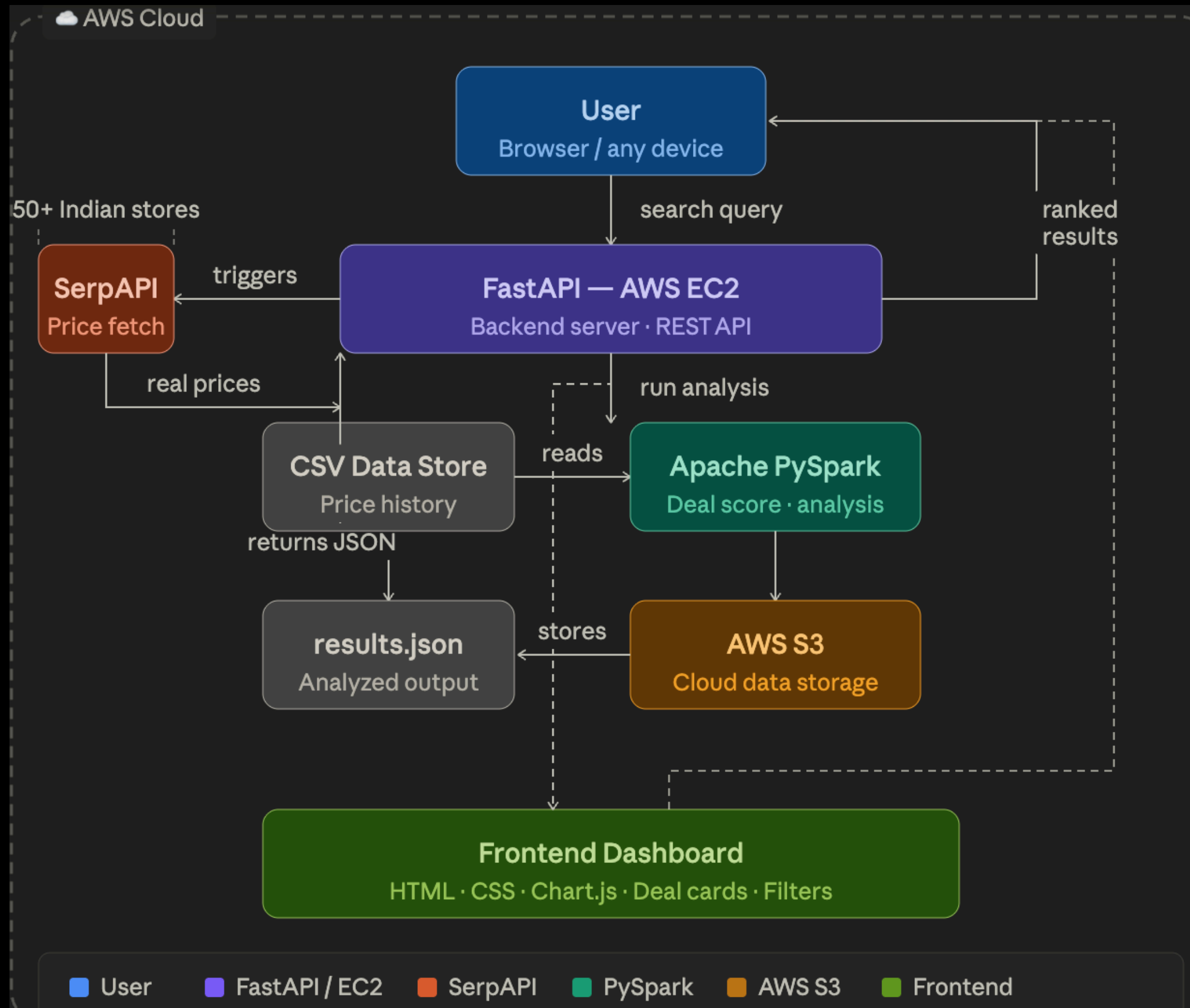
Savings Calculator

Enter quantity, instantly see total cost at best price and exact rupees saved vs market high."

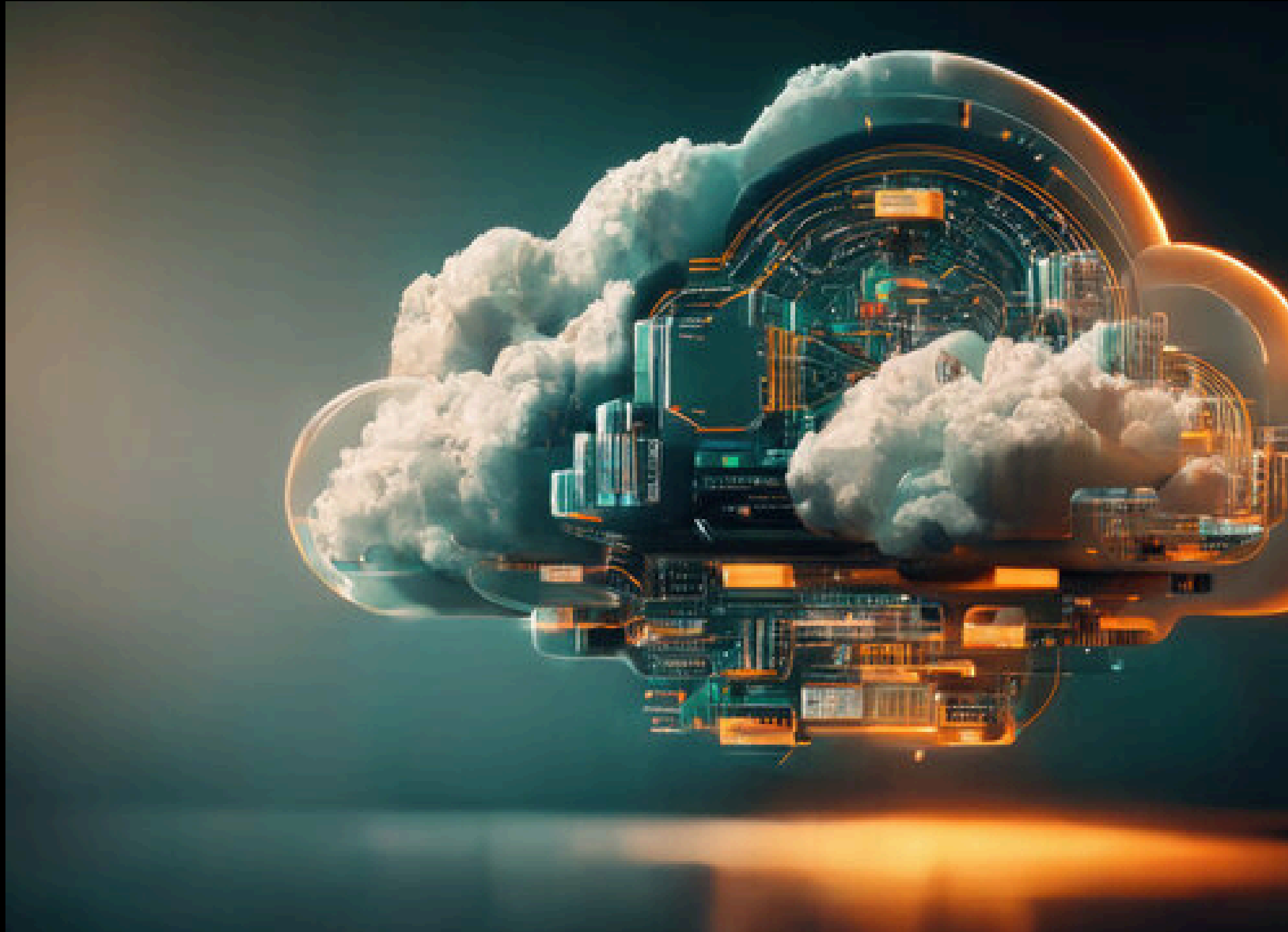
Technologies Used

Layer	Technology
Data Fetching	SerpAPI + Python
Big Data Processing	Apache PySpark
Backend API	FastAPI
Frontend	HTML, CSS, JavaScript, Chart.js
Cloud Deployment	AWS EC2, S3

Architecture



AWS Stack Overview



Cloud Platform

AWS (Amazon Web Services)

Server

EC2 t2.micro instance — Amazon Linux 2023

Storage

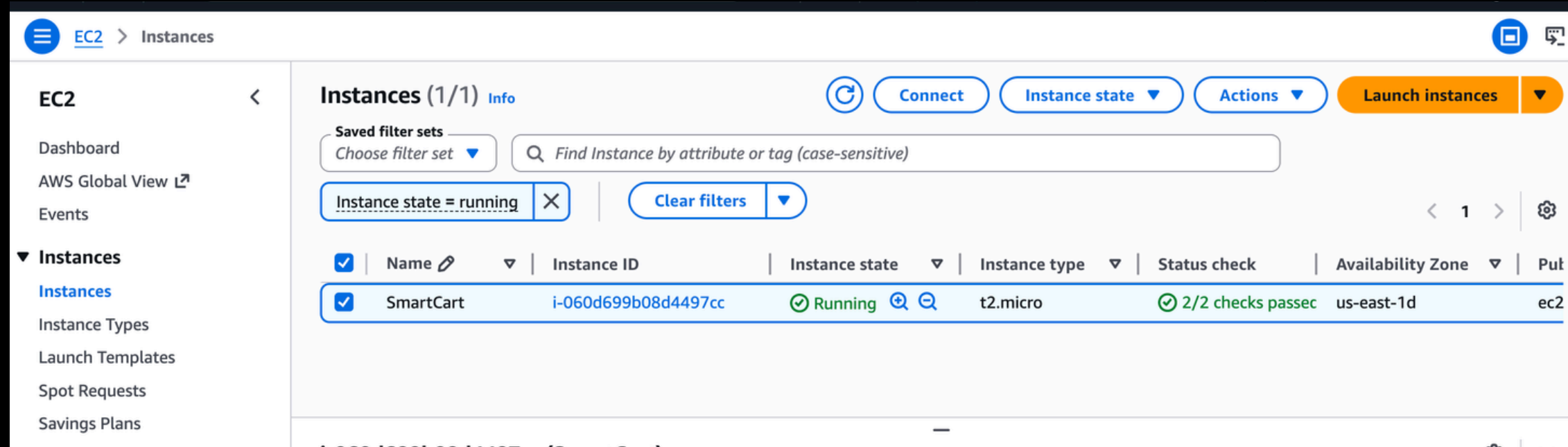
AWS S3 bucket for price data

Network Access

Public IPv4 — accessible from any browser worldwide

Deployment Flow

Launch EC2 instance on AWS



Console SSH into server from local terminal

```
[(base) arushi@Arushikas-MacBook-Air ~ % ssh -i ~/Downloads/smartcart-key.pem ec2-user@34.224.72.53
The authenticity of host '34.224.72.53 (34.224.72.53)' can't be established.
ED25519 key fingerprint is: SHA256:ITKvycPONOHq6Qcr5UclLIqdsEPo0lQeD5NmDhnSRJU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '34.224.72.53' (ED25519) to the list of known hosts.

      #_
    ~\_ #####_      Amazon Linux 2023
  ~~  \_#####\
  ~~   \###|
  ~~    \#/  ___  https://aws.amazon.com/linux/amazon-linux-2023
  ~~      V~'  '→
    ~~~
      ~~.~.~
        _/_/_/_/_
        _/m/'

[ec2-user@ip-172-31-23-82 ~]$ Connection to 34.224.72.53 closed by remote host.
```


Install Python, Java 17, PySpark, FastAPI

```
[ec2-user@ip-172-31-23-82 ~]$ cd smartcart
[ec2-user@ip-172-31-23-82 smartcart]$ ls
api fetch_prices.py frontend pyspark_scripts requirements.txt
[ec2-user@ip-172-31-23-82 smartcart]$ pip3 install fastapi uvicorn python-dotenv requests aiofiles jinja2 python-multipart boto3
Defaulting to user installation because normal site-packages is not writeable
Collecting fastapi
  Using cached fastapi-0.128.8-py3-none-any.whl (183 kB)
Collecting uvicorn
  Using cached uvicorn-0.39.0-py3-none-any.whl (68 kB)
Collecting python-dotenv
  Using cached python_dotenv-1.2.1-py3-none-any.whl (21 kB)
Requirement already satisfied: requests in /usr/lib/python3.9/site-packages (2.28.1)
Collecting aiofiles
  Downloading aiofiles-25.1.0-py3-none-any.whl (14 kB)
Requirement already satisfied: jinja2 in /usr/lib/python3.9/site-packages (2.11.3)
Collecting python-multipart
  Downloading python_multipart-0.0.20-py3-none-any.whl (24 kB)
Collecting boto3
  Downloading boto3-1.42.97-py3-none-any.whl (140 kB)
  |████████████████████| 140 kB 24.0 MB/s
Collecting pydantic>=2.7.0
  Downloading pydantic-2.13.4-py3-none-any.whl (472 kB)
  |████████████████████| 472 kB 38.2 MB/s
Collecting starlette<1.0.0,>=0.40.0
  Downloading starlette-0.49.3-py3-none-any.whl (74 kB)
  |████████████████████| 74 kB 6.4 MB/s
```

Upload project via SCP

```
ushu@Arushikas-MacBook-Air ~ % scp -i ~/Downloads/smartcart-key.pem ~/Download(base) arushi@Arushikas-MacBook-Air ~ % scp -i ~/Downloads/smartcart-key.pem ~/Download(base) arushi@Arushikas-
p -i ~/Downloads/smartcart-key.pem ~/Downloads/frontend/index.html ec2-user@3.88.216.117:/home/ec2-user/smartcart/frontend/index.html
ushu@Arushikas-MacBook-Air ~ % scp -i ~/Downloads/smartcart-key.pem ~/Downloads/analyze.py ec2-user@3.88.216.117:/home/ec2-user/smartcart/pyspark_scripts/analyze.py
y
100% 5447 11.6KB/s 00:00
ushu@Arushikas-MacBook-Air ~ % scp -i ~/Downloads/smartcart-key.pem ~/smartcart/fetch_prices.py ec2-user@3.88.216.117:/home/ec2-user/smartcart/fetch_prices.py
ces.py
100% 9341 18.4KB/s 00:00
ushu@Arushikas-MacBook-Air ~ % scp -i ~/Downloads/smartcart-key.pem ~/smartcart/frontend/index.html ec2-user@3.88.216.117:/home/ec2-user/smartcart/frontend/index.html
l
100% 104KB 164.4KB/s 00:00
ushu@Arushikas-MacBook-Air ~ % ssh -i ~/Downloads/smartcart-key.pem ec2-user@3.88.216.117 "pkill -f uvicorn; pkill -f python; cd smartcart && export JAVA_HOME=/usr/lib/jvm/java-17-amazon-co
ATH=$JAVA_HOME/bin:$PATH && nohup uvicorn api.main:app --host 0.0.0.0 --port 8000 &"
ushu@Arushikas-MacBook-Air ~ %
```

Configure JAVA_HOME environment variable
Start server with nohup uvicorn — runs 24/7
App live at http://EC2-IP:8000

```

    ~~~
    ~~. _ .
    _/ _/ _/
    _/m/ '
Last login: Tue May 26 12:44:08 2026 from 104.28.157.138
[ec2-user@ip-172-31-23-82 ~]$ cd smartcart
export JAVA_HOME=/usr/lib/jvm/java-17-amazon-corretto
export PATH=$JAVA_HOME/bin:$PATH
nohup uvicorn api.main:app --host 0.0.0.0 --port 8000 > server.log 2>&1 &

```

AWS S3 bucket with data files

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems New

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

smartcart-project-29 Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (2)

Copy S3 URI

Copy URL

Download

Open ↗

Delete

Actions ▼

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

<

1

>

☐	Name	Type	Last modified	Size	Storage class
☐	data/	Folder	-	-	-
☐	frontend/	Folder	-	-	-

THANK YOU!